

A Real-Time Microstructure Analytics Engine and Gated Large Language Model Synthesis Daemon for High-Conviction Bitcoin Volatility Discovery

Abstract

Modern algorithmic cryptocurrency markets exhibit severe signal degradation caused by fragmented liquidity, high-frequency synthetic positioning, and reflexive order flow loops. Standard momentum and price-action technical indicators regularly generate false positives during range-bound rotational regimes, failing to distinguish between genuine spot accumulation and leveraged derivative short squeezes. This paper presents the architecture, theoretical thesis, and operational framework of btc-alert-system, a high-throughput, real-time Bitcoin microstructure analytics daemon. The system ingests dual tick feeds from spot (Binance) and perpetual derivative (Bybit) protocols, dynamically calculates rolling Cumulative Volume Delta (CVD) divergence and discretized Volume Profiles (Point of Control, Value Area High/Low), and applies a zero-cost deterministic state engine. Under verified structural acceptance outside value areas, the daemon deploys an event-gated, schema-constrained Large Language Model (gemini-3.6-flash) reasoning layer to evaluate market conviction and dispatch actionable, low-noise alerts via WhatsApp.

1. Introduction

Cryptocurrency price discovery occurs primarily across two distinct yet deeply coupled venues: the spot exchange market and the perpetual futures market. Because derivative contracts allow substantial financial leverage, directional expansion on perpetual markets frequently manifests without underlying physical asset accumulation. Such moves remain vulnerable to abrupt exhaustion and cascading long/short liquidation events.

Conversely, sustainable structural breakouts require institutional spot absorption—the permanent removal of floating supply from the order books. Traditional technical analysis frameworks based on static price indicators (e.g., RSI, moving averages) process price as an unweighted scalar, discarding the underlying trade size, aggressor side, and venue classification.

btc-alert-system addresses this analytical gap by treating market microstructure as a multi-dimensional state space. By operating directly on raw WebSocket trade-level tick data (T), the daemon isolates market conviction through order-flow delta divergence and structural auction theory, converting quantitative market anomalies into natural language intelligence.

2. Theoretical Thesis and Quantitative Foundation

The core operational thesis of btc-alert-system is grounded in **Auction Market Theory (AMT)** and **Aggressor Order Flow Dynamics**.

2.1. Dual-Venue Cumulative Volume Delta (CVD)

Every executed trade tick i contains price p_i , quantity q_i , and an aggressor flag $a_i \in \{-1, +1\}$ indicating whether the trade matched against a resting limit bid (-1 , sell aggressor) or ask ($+1$, buy aggressor). The rolling tick delta $\delta_i = a_i \cdot q_i$ is computed

independently for spot ($\mathcal{S}$) and perpetual ($\mathcal{P}$) venues over a sliding temporal window $\Delta t = 60$ minutes:

$$\text{CVD}_{\mathcal{S}}(t) = \sum_{t_i \in [t-\Delta t, t]} a_i^{\mathcal{S}} q_i^{\mathcal{S}}, \quad \text{CVD}_{\mathcal{P}}(t) = \sum_{t_i \in [t-\Delta t, t]} a_i^{\mathcal{P}} q_i^{\mathcal{P}}$$

The **CVD Divergence Metric** (Δ_{CVD}) measures the structural disparity between spot buyers and perpetual speculators:

$$\Delta_{\text{CVD}}(t) = \text{CVD}_{\mathcal{S}}(t) - \text{CVD}_{\mathcal{P}}(t)$$

- $\Delta_{\text{CVD}} \gg 0$ **during upward price drift**: Spot-Led Expansion. Spot participants are actively consuming ask liquidity, confirming institutional accumulation.
- $\Delta_{\text{CVD}} \ll 0$ **during upward price drift**: Leveraged Speculation / Squeeze Risk. The upward drift is driven almost entirely by derivative leverage with spot absorption absent, presenting high risk for long liquidations back into the mean.

2.2. Rolling Volume Profile and Auction Distribution

Price levels are discretized into uniform bins of size $B = \$25.00$. Total transacted volume is mapped to a discrete probability density function:

$$V(b) = \sum_{p_i \in [b, b+B)} q_i$$

From this distribution, the daemon derives:

1. **Point of Control (POC)**: The price bin possessing maximum traded density:

$$\text{POC} = \arg \max_b V(b)$$

2. **Value Area (VA)**: The smallest contiguous set of price bins containing 70% ($\alpha = 0.70$) of total window volume:

$$\sum_{b \in [\text{VAL}, \text{VAH}]} V(b) = \alpha \sum_{\forall b} V(b)$$

where VAH and VAL denote the Value Area High and Value Area Low boundaries, respectively.

2.3. Hybrid Gated Reasoning Architecture

Continuous API polling of Large Language Models introduces latency and rapidly exhausts finite rate limits. To maintain zero cost during routine market conditions while preserving deep qualitative synthesis for major anomalies, btc-alert-system uses a dual-engine architecture:

1. **Deterministic Rule Engine (99% State):** If $p(t) \in [VAL, VAH]$ or total rolling volume $V_{total} < 50$ BTC, the daemon assigns deterministic, pre-scripted state classifications (Mean Reverting / Consolidation) with zero API consumption.
2. **LLM Inference Gate (1% State):** Gemini (gemini-3.6-flash) is invoked strictly when all criteria align:
 - **Boundary Violation:** $p(t) > VAH$ or $p(t) < VAL$
 - **Volume Validation:** $V_{total} \geq 100$ BTC with minimum warmup period $t_{warmup} \geq 300s$
 - **Delta Divergence:** $|\Delta_{CVD}| \geq 25$ BTC
 - **Temporal Cooldown:** $\Delta t_{inference} \geq 900s$ (15 minutes)

The reasoning engine returns a schema-constrained Pydantic structure (MicrostructureAnalysis) enforcing strict typings for regime, uncertainty_level, verbal_summary, and key_risk_factor.

3. System Architecture & Technical Specifications

Component	Technology	Functional Specification
Ingestion Layer	websockets, asyncio	Concurrent async ingestion of Binance Spot (aggTrade) and Bybit Linear (publicTrade.BTCUSDT)
Analytics Engine	collections.deque, numpy	$O(1)$ amortized tick insertion and rolling window metric pruning over a 60-minute horizon
Reasoning Engine	google-genai (gemini-3.6-flash)	Schema-enforced structured JSON output with Automatic Function Calling disabled
Terminal UI	rich.live, rich.layout	Real-time multi-pane dashboard displaying metrics, volume levels, and qualitative commentary
Alert Bridge	aiohttp, CallMeBot API	HTTPS asynchronous dispatch to encrypted WhatsApp endpoint with 30-minute deduplication lock

Component	Technology	Functional Specification
Process Management	tmux, Linux POSIX	Detached background daemonization on local Linux / WSL workstations

4. Operational Guide: Post-Deployment Execution

Once deployed, btc-alert-system is designed to run autonomously as a background monitor.

4.1. Execution & Session Management

To manage the lifecycle of the daemon without tying up the active terminal shell:

- **Starting the Daemon:** Execute `./run_daemon.sh` from the repository root. This initializes the Python virtual environment and detaches the process inside a dedicated tmux session named `btc-daemon`.
- **Attaching to the Workstation UI:** Run `tmux attach -t btc-daemon` to view the live Rich layout, including real-time CVD indicators, tick frequencies, POC boundaries, and Gemini analytical summaries.
- **Detaching from UI:** Press `Ctrl + B`, release both keys, and press `D`. The terminal will detach while the daemon continues streaming and calculating metrics in the background.
- **Terminating the Engine:** Attach to the session and send a SIGINT via `Ctrl + C`, or execute `tmux kill-session -t btc-daemon`.

4.2. Interpreting Alert Signals

The system will dispatch notifications to WhatsApp strictly when institutional conviction is established.

When an alert is received:

1. **Verify Venue Leadership:** Confirm that Δ_{CVD} aligns with the price breakout direction. A breakout above VAH accompanied by positive spot CVD represents genuine institutional absorption, whereas a breakout accompanied by negative spot CVD indicates an impending leverage squeeze.
2. **Monitor Value Area Acceptance:** If price re-enters the Value Area and breaks below the POC, the breakout has failed, indicating mean-reversion toward the opposing boundary (VAL).
3. **Assess Cooldown Boundaries:** The daemon enforces a 30-minute lock after an alert. Re-attach to the TUI to monitor tick accumulations and order book migration across subsequent value area shifts.

5. Conclusion

btc-alert-system bridges quantitative microstructure order flow analytics and semantic artificial intelligence. By gating LLM inference behind strict structural and volume divergence thresholds, the architecture eliminates noise, avoids API rate limits, and provides unambiguous, high-conviction Bitcoin breakout discovery.